

Competitive Programming

Section 46 Editorial: Rerooting and Advanced Tree DP

Competitive Programming Course

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1 Editorial overview

1.1 Problem-specific proof map

A. Sum Distances. First compute the root answer and subtree sizes. Moving the root across parent-child edge (u, v) removes $\text{size}[v]$ near vertices and adds $n - \text{size}[v]$ far vertices, giving $\text{ans}[v] = \text{ans}[u] + n - 2 * \text{size}[v]$.

B. Farthest Node Distance. The farthest distance through a child uses that child's downward value; the farthest outside it uses the best sibling or parent value. Prefix/suffix maxima exclude the child without double counting.

C. Paths Through Node. Removing a node splits the tree into component sizes. Every path through it chooses endpoints from two different components (or the node), so summing cross products counts each unordered pair exactly once.

1.2 Detailed solution guide

A. Sum of Distances from Every Root. First root the tree at 1. A postorder DFS computes $\text{sub}[v]$ and $\text{down}[v]$, where $\text{down}[v]$ is the sum of distances from v to vertices in its subtree. The answer for root 1 is $\text{down}[1]$. When rerooting across edge $v - c$, the $\text{sub}[c]$ vertices inside c 's subtree become one step closer, and the other $n - \text{sub}[c]$ vertices become one step farther. Therefore $\text{ans}[c] = \text{ans}[v] - \text{sub}[c] + (n - \text{sub}[c])$.

B. Tree Diameter. Run a DFS/BFS from any vertex to find a farthest vertex a . Run again from a ; the farthest vertex b gives the diameter distance. The first search reaches an endpoint of some diameter because any path can be extended toward a farthest vertex, and the second search measures the longest path starting from that endpoint.

C. Pairs Separated by Removing a Vertex. Removing vertex v splits the tree into components: one for each child subtree and, unless v is the root, one parent-side component of size $n - \text{sub}[v]$. Pairs whose path passes through v are exactly pairs chosen from two different components. Summing component sizes seen so far gives $\text{sum} += \text{seen} * \text{comp_size}$, then $\text{seen} += \text{comp_size}$.

Common pitfalls. In A, the reroot formula changes distances for all vertices, not just descendants. In C, decide whether pairs are ordered or unordered before multiplying by two; the local task uses unordered pairs.

Each task asks for every-node information on a tree.

- A first computes subtree sizes and the answer for root 1; moving the root across edge $u - v$ changes distances by $-\text{sub}[v] + (n - \text{sub}[v])$.
- B uses the diameter endpoint theorem: the farthest node from any vertex is one of the two diameter endpoints.
- C removes each node conceptually and counts pairs of endpoints lying in different resulting components.

The proof technique is contribution accounting: decide how one edge/root move changes every affected node or pair exactly once.

2 A. Sum Distances

2.1 Restatement

For every node, print the sum of distances to all other nodes.

2.2 Algorithm and proof

The task-specific reduction in the proof map maps legal original choices to exactly the generated states, and the final operation reads the requested optimum or classification. \square

2.3 Complexity

The bound is the concrete one stated for this task in the proof map; the code performs only its enumerations, searches, or transitions.

2.4 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std; int main(){ios::sync_with_stdio(false);cin.tie(nullptr);int n;if(!(cin>
```

2.5 Full Python solution

```
import sys
sys.setrecursionlimit(1_000_000)
def main():
    data=list(map(int,sys.stdin.buffer.read().split()))
    if not data: return
    n=data[0]; g=[[] for _ in range(n)]; idx=1
    for _ in range(n-1):
        a,b=data[idx]-1,data[idx+1]-1; idx+=2; g[a].append(b); g[b].append(a)
    sub=[1]*n; down=[0]*n
    def dfs(u,p):
        for v in g[u]:
            if v!=p: dfs(v,u); sub[u]+=sub[v]; down[u]+=down[v]+sub[v]
    ans=[0]*n
    def reroot(u,p):
```

```

    for v in g[u]:
        if v!=p:
            ans[v]=ans[u]+n-2*sub[v]; reroot(v,u)
    dfs(0,-1); ans[0]=down[0]; reroot(0,-1); print(" ".join(map(str,ans)))
if __name__=="__main__": main()

```

3 B. Farthest Node Distance

3.1 Restatement

For every node, print its distance to the farthest node.

3.2 Algorithm and proof

The task-specific reduction in the proof map maps legal original choices to exactly the generated states, and the final operation reads the requested optimum or classification. \square

3.3 Complexity

The bound is the concrete one stated for this task in the proof map; the code performs only its enumerations, searches, or transitions.

3.4 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std; vector<int>bfs(int s,vector<vector<int>>&g){vector<int>d(g.size(),-1);c
```

3.5 Full Python solution

```
import sys,collections
def bfs(s,g):
    d=[-1]*len(g); d[s]=0; q=collections.deque([s])
    while q:
        u=q.popleft()
        for v in g[u]:
            if d[v]<0: d[v]=d[u]+1; q.append(v)
    return d
def main():
    data=list(map(int,sys.stdin.buffer.read().split()))
    if not data: return
    n=data[0]; g=[[] for _ in range(n)]; idx=1
    for _ in range(n-1):
        a,b=data[idx]-1,data[idx+1]-1; idx+=2; g[a].append(b); g[b].append(a)
```

```
    d0=bfs(0,g); a=max(range(n),key=d0.__getitem__); da=bfs(a,g); b=max(range(n),key=da.__getitem__)
    print(" ".join(str(max(da[i],db[i])) for i in range(n)))
if __name__=="__main__": main()
```

4 C. Paths Through Node

4.1 Restatement

For every node, count unordered pairs of distinct nodes whose path passes through it.

4.2 Algorithm and proof

The task-specific reduction in the proof map maps legal original choices to exactly the generated states, and the final operation reads the requested optimum or classification. \square

4.3 Complexity

The bound is the concrete one stated for this task in the proof map; the code performs only its enumerations, searches, or transitions.

4.4 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std; int main(){ios::sync_with_stdio(false);cin.tie(nullptr);int n;if(!(cin>
```

4.5 Full Python solution

```
import sys
sys.setrecursionlimit(1_000_000)
def main():
    data=list(map(int,sys.stdin.buffer.read().split()))
    if not data: return
    n=data[0]; g=[[] for _ in range(n)]; idx=1
    for _ in range(n-1):
        a,b=data[idx]-1,data[idx+1]-1; idx+=2; g[a].append(b); g[b].append(a)
    parent=[-1]*n; order=[0]
    for u in order:
        for v in g[u]:
            if v!=parent[u]: parent[v]=u; order.append(v)
    sub=[1]*n
    for u in reversed(order[1:]): sub[parent[u]]+=sub[u]
```

```
total=n*(n-1)//2; out=[]
for u in range(n):
    bad=0
    for v in g[u]:
        s=sub[v] if parent[v]==u else n-sub[u]
        bad+=s*(s-1)//2
    out.append(str(total-bad))
print(" ".join(out))
if __name__=="__main__": main()
```

5 Testing discussion

For every reroot transition, classify which vertices become closer and farther; this is the correctness proof. Brute-force BFS from every root for random trees of at most eight vertices. For path-through-node counts, remove the node conceptually: pairs in different resulting components contribute, while pairs inside one component do not.

Random tests compare C++ and Python against fixed expected outputs generated from the reference implementation. Important cases include singletons, negative numbers where allowed, repeated values, and boundary query sizes.